

Rout Qulla  
1010683

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classmate

Date

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SP 104

## Machine Learning.

$$(1) \quad A \in \{0, 1\}.$$

$$B \in \{0, 1, 2\}.$$

$$P(A=0, B=0) = \frac{1}{24} (P_{00}).$$

$$P(A=0, B=1) = \frac{1}{12} (P_{01}).$$

$$P(A=0, B=2) = \frac{1}{8} (P_{02}).$$

$$P(A=1, B=0) = \frac{1}{8} (P_{10}).$$

$$P(A=1, B=1) = \frac{1}{4} (P_{11}).$$

$$P(A=1, B=2) = P_{12}.$$

$$(a) \quad \sum P_{ij} = 1. \quad (\text{condition for JOINT PROBABILITY}).$$

$$\frac{1}{24} + \frac{1}{12} + \left(\frac{1}{8} + \frac{1}{8}\right) + \frac{1}{4} + P_{12} = 1.$$

$$\Rightarrow \frac{1}{24} + \frac{1}{12} + \frac{1}{2} + P_{12} = 1.$$

$$\Rightarrow P_{12} + \frac{1+2+2+1}{24} = 1.$$

$$\Rightarrow P_{12} = \frac{24 - 12 - 2 - 1}{24} = \frac{9}{24} = \frac{3}{8}.$$

$$\Rightarrow \boxed{P(A=1, B=2) = \frac{3}{8}}.$$



$$(b) \quad P(B=0) = P(B=0, A=1) + P(B=0, A=0) \\ = \frac{1}{24} + \frac{1}{8} = \frac{4}{24} = \frac{1}{6}$$

$$P(B=1) = P(A=0, B=1) + P(A=1, B=1) \\ = \frac{1}{12} + \frac{1}{4} = \frac{4}{12} = \frac{1}{3}$$

$$P(B=2) = P(A=0, B=2) + P(A=1, B=2) \\ = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

(c) Expectation (for discrete variable B).

$$\rightarrow \sum_{x \in B \subseteq \{0, 1, 2\}} x \cdot P(x) = 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{2} \\ = 1 + \frac{1}{3} = \frac{4}{3}$$

Variance (for discrete variable B).

$$\rightarrow \sum x^2 \cdot P(x) - \left[ \sum x \cdot P(x) \right]^2 \\ \Rightarrow \left( 0^2 \cdot \frac{1}{6} + 1^2 \cdot \frac{1}{3} + 2^2 \cdot \frac{1}{2} \right) - \left( \frac{4}{3} \right)^2 \\ = \frac{1}{3} + 2 - \frac{16}{9} = \frac{3}{9} + \frac{18}{9} - \frac{16}{9} = \frac{5}{9}$$



(a) Marginal distribution of A.

$$P(A=0) = P(\cancel{A=0, B})$$

$$P_{00} + P_{01} + P_{02} = \frac{1}{24} + \frac{1}{12} + \frac{1}{4} =$$

$$= \frac{3+2+1}{24} = \frac{6}{24} = \left(\frac{1}{4}\right)$$

$$P(A=1) = P_{10} + P_{11} + P_{12} = \frac{1}{8} + \frac{1}{4} + \frac{3}{8}$$

$$= \frac{4}{8} + \frac{2}{8} = \frac{6}{8} = \left(\frac{3}{4}\right)$$

For a joint distribution, the condition for both of the random variables to be independent, the marginal dists must be COMPLETE, and the JOINT DISTs should be products of individual MARGINAL DISTs.

$$P(A=0, B=2) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8} = P(A=0) \times P(B=2)$$

$$\rightarrow P(A=i, B=j) = P(A=i) \cdot P(B=j) \quad \forall i, j \in A, B$$

$\therefore A$  &  $B$  are independent variables.



$$1.) P(X=x) = (1-\theta)^n \cdot \theta^x \quad [0 \in (0,1)]$$

(4)  $n$  are discrete ~~partia~~ integers.

$$P(x_1) \cdot P(x_2) \cdot P(x_3) \cdots P(x_n)$$

$$\text{MLE} \Rightarrow \prod_{i=1}^n (1-\theta) \theta^{x_i} = (1-\theta)^n \theta^{\sum_{i=1}^n x_i}$$

$$\log(\text{MLE}) = n \log(1-\theta) + \log(\theta) \left( \sum_{i=1}^n x_i \right)$$

$$\frac{\partial}{\partial \theta} \log(\text{MLE}) = \frac{-n}{1-\theta} + \frac{\sum_{i=1}^n x_i}{\theta} = 0$$

$$\frac{\sum_{i=1}^n x_i}{1-\theta} = \frac{\sum_{i=1}^n x_i}{\theta}$$

$$\Rightarrow \frac{1}{1-\theta} = \frac{\theta}{\sum_{i=1}^n x_i}$$

$$\Rightarrow \theta \left( \frac{1}{1-\theta} + \frac{1}{\sum_{i=1}^n x_i} \right) = \frac{1}{\frac{1}{1-\theta} + \frac{1}{\sum_{i=1}^n x_i}}$$

$$\theta = \frac{1}{1 + \frac{n}{\sum_{i=1}^n x_i}} = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i + n} \quad (\text{ANS})$$

ANS  $\rightarrow \hat{\theta} = \frac{\sum_{i=1}^n x_i}{\sum_{i=1}^n x_i + n}$  is the MLE

(8)  $A \geq \frac{1}{2}$  (equal probability of small/big)

final definitely small.  $\rightarrow$  initial (small/big)

GALIA BOX



First removed  
 $P[\text{Initial item being small}]$

$= P(\text{second item chosen}) \times P(\text{second item being small})$

$+ P(\text{first item chosen}) \times P(\text{first item being small})$

$$= \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\therefore B = \frac{3}{4}$$

$P(\text{remaining item is small})$

$= P(\text{initial item chosen}) \cdot P(\text{final item is small})$

$+ P(\text{final item chosen}) \cdot P(\text{initial item is small})$

$$= \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} \therefore C = \frac{3}{4}$$

Question: What is the probability that the remaining item is small given that the first item was definitely small.  $= P(C|B)$



$$\cancel{P(C|B)} = \frac{\cancel{P(C)}}{\cancel{P(B)}} = \frac{\cancel{3/4}}{\cancel{3/4}} = 1$$

$$P(C|B) = \frac{P(C \cap B)}{P(B)} \quad (\text{Baye's theorem / conditional probability law})$$

$\therefore P(C \cap B) =$  first item is small AND remaining item is also small.

$$= \frac{1}{2} \left( \frac{1}{2} \cdot 1 \right) + \frac{1}{2} \left( 1 \cdot \frac{1}{2} \right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

first item (small)      second item (small)      first item (small)      second item (small)

$$\therefore P(C|B) = \frac{P(B \cap C)}{P(B)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$$

$$(3) \quad \mu_1 = \frac{2+8}{\sum_{i=1}^n i} = \frac{2+8}{2} = 1+4 = 5$$

$$(2) \quad \mu_2 = \frac{1+3+9+10}{\sum_{i=1}^n i} = \frac{23}{4} = 5.75$$



(E) Nearest centroids to each pt.

(\*) Since there are two centroids we MAY have 2 clusters.

$$① \min(|1-5|, |1-5.75|) = 4 (D_1)$$

$$② \min(|2-5|, |2-5.75|) = 3 (D_1)$$

$$③ \min(|3-5|, |3-5.75|) = 2 (D_1)$$

$$④ \min(|8-5|, |8-5.75|) = 2.25 (D_2)$$

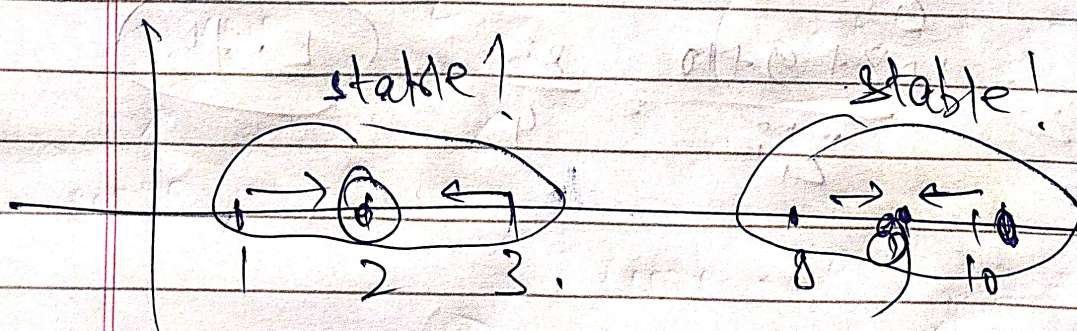
$$⑤ \min(|9-5|, |9-5.75|) = 3.25 (D_2)$$

$$⑥ \min(|10-5|, |10-5.75|) = 4.25 (D_2)$$

New centroids,  $\mu_1^* = \frac{1+3+2}{3} = 2$

$$\mu_2^* = \frac{8+9+10}{3} = 9$$

$$D_1 = \{1, 2, 3\}, D_2 = \{8, 9, 10\}$$





(4.) (a).  $f(x, y) = \begin{cases} k x^3 y^2 & \text{if } x \in [0, 1], y \in [0, 2]. \\ 0 & \text{if } x \notin [0, 1], y \notin [0, 2]. \end{cases}$

$$\therefore \int_0^1 dx \int_0^2 dy k x^3 y^2 = 1. \quad [\text{Probability law}]$$

$$\Rightarrow k \left[ \frac{x^4}{4} \right]_0^1 \left[ \frac{y^3}{3} \right]_0^2 \Rightarrow k \cdot \frac{1}{4} \cdot \frac{8}{3} = 1$$

$$\Rightarrow \boxed{k = \frac{3}{2}}$$

$$\therefore \text{pdf}_{(x,y)} \Rightarrow \frac{3}{2} x^3 y^2 \quad \forall x \in [0, 1], y \in [0, 2].$$

(b) pdf(x)  $\Rightarrow$  marginal probability of x.

$$\Rightarrow \frac{3}{2} x^3 \int_0^2 y^2 dy = \frac{3}{2} x^3 \cdot \left[ \frac{y^3}{3} \right]_0^2 = 4x^3$$

$$\therefore \text{pdf}(x) = 4x^3$$

(c) Multivariate function:  $\frac{3}{2} x^3 y^2$ .

We have to find the region where  $(y < x)$  in order to find  $P(X > Y)$ .



We must remember  $y \in [0, 2]$ .  
 $\therefore$  for  $x > y$ ,  $y \in [0, 1]$ .

Let's take the  $y$  component.

$$\phi(y) = y^2. \quad \therefore \phi(y|x) = \int_y^x y^2 dy$$

$$= \left[ \frac{y^3}{3} \right]_y^x = \frac{x^3}{3} - \frac{y^3}{3}$$

Total area of TMAP

$$= \int_0^1 \frac{1}{2} x^3 \cdot \frac{x^3}{2} dx = \frac{1}{2} \int_0^1 x^6 dx = \frac{1}{2} \left[ \frac{x^7}{7} \right]_0^1 = \frac{1}{14}$$

### (5) RIASEC code book.

It's a regression question and to check the variation of  $R_i$  parameter with that of  $R_{score}$ .

$$R_{score} = \frac{\sum_{i=1}^8 R_i}{\sum_{i=1}^8 1} = \frac{R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 + R_8}{8}$$

$R_{score}$

$$R_{score} = \alpha R_i + \beta$$